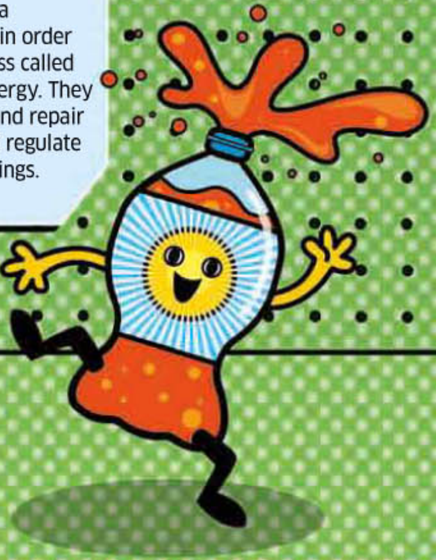
BIOLOGIST BELIEFS

Biology is the study of living things, so biologists certainly have an opinion here. This question has been bugging them since life began (practically), but there are probably as many opinions as there are biologists. Still, they have agreed that there is a sort of “shopping list” of traits common to all living things. Without checking all of these boxes, there is no life. You’ve checked out.

ENERGY

Living things need to take in a continuous supply of energy in order to function. They use a process called metabolism to access that energy. They need energy to move, grow and repair cells, transport nutrients, and regulate temperature, among other things.

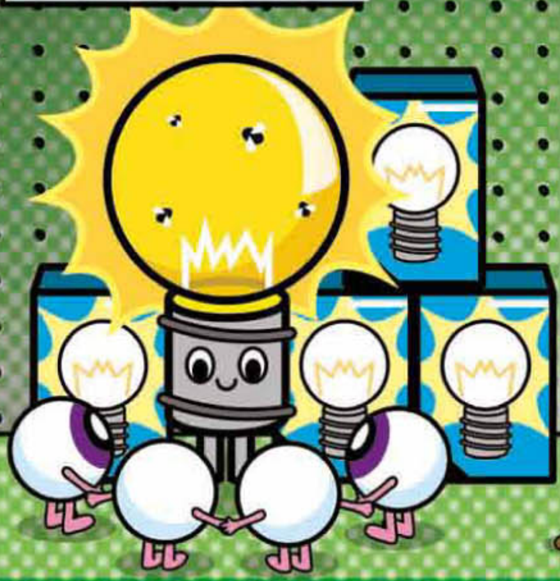


CELLS

All living things require a cell or cells. Cells are the basic ingredients of life, and the smallest units of living matter. Every single living organism on Earth is divided into pieces called cells, each with its own set of functions. There are trillions of cells inside you.

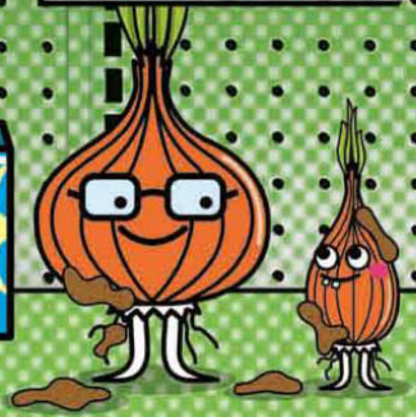
RESPONSIVENESS

Another trait common to living things is responsiveness to their environment. For example, a living thing will respond to a change in sound, heat, light, or contact. Even a simple single-celled critter can detect and move toward light.



REPRODUCTION

A living thing can make a copy of itself through reproduction. Some organisms can reproduce by themselves, which is called asexual reproduction. Others reproduce in pairs—sexual reproduction. They pass their traits to their offspring.



EVOLUTION

Over time, living organisms have the ability to slowly change, or evolve, to better adapt to their environment. They develop ways to cope with their surroundings that their ancestors did not have, so they can live better.

VIRUS ALERT

Although there is plenty of debate, most biologists do not think that viruses are alive. They evolve, have genes, and replicate by making copies of themselves. But viruses are not made of cells (they are little more than a strand of DNA covered with protein), do not have an independent metabolism, and do not take in energy, so they are not considered alive. Even so, they are nothing to sneeze at once they get in your system.

COMPUTER VIRUS

A computer virus is a small piece of rogue software that attaches itself onto real programs and can wreak havoc on a computer. Some people speculate that computer viruses are alive, as they reproduce by attaching to other programs. Although they pass from machine to machine to infect it like a virus spreads between people, they copy themselves exactly, with no possibility of evolution . . . yet. So, a computer virus cannot really be considered alive.